ECE343 Spring 2025

Objective 3.4: Impedance Matching



Lesson	Topics	Reading	Notes
1	Course Overview & Introduction	Ch.1	LO 1.1-1.2
Chapter 2: Electric and Magnetic Fields			
2	Fields, Parameters, Field Quantities	Ch. 2	LO 2.1-2.2
Chapter 3: Transmission Lines			
3	Lumped Element Model	3.1-3.4	LO 3.1
4	Telegrapher's Equations	3.5-3.6	LO 3.1
5	Characteristic Impedance & Wave Propagation	3.7-3.9	LO 3.2
6	Coaxial, Reflection Coefficient, VSWR	3.10,3.12-3.14	LO 3.2-3.3
7	Terminated Transmission Lines	3.15,3.17-3.19	LO 3.4
8	Impedance Matching	3.21-3.23	LO 3.4
9	Impedance Matching	3.21-3.23	LO 3.4
10	Smith Charts	Supplemental	LO 3.5
11	Smith Charts	Supplemental	LO 3.5
12	Smith Charts	Supplemental	LO 3.5
13	Smith Charts	Supplemental	LO 3.5
14	Transients	Supplemental	LO 3.6
15	Reflection/Transmission Coefficient Lab	Lab Packet	LO 3.7
Chapter 4: Vector Analysis			
16	Vector & Coordinate System Review	Ch. 4	LO 4.1-4.4 **GR 1 due**
Chapter 5: Electrostatics			
17	Coulomb's Law	5.1-5.3	LO 5.1
18	Surface and Volume Charge	5.4	LO 5.1
19	Gauss' Law (Integral Form)	5.5-5.6	LO 5.2
20	Gauss' Law (Differential Form)	5.7-5.8	LO 5.2
21	Electric Potential	5.9-5.10, 5.15	LO 5.3

Ch 3 Learning Objectives

3.1 I can describe the origin of the transmission line equation, understand the method of developing a solution, and interpret the solution.

3.2 I can explain the physical meaning of and calculate the following quantities: propagation constant, phase constant, attenuation constant, characteristic impedance, VSWR, reflection coefficient, and transmission coefficient.

3.3 I can use the transmission line equation to calculate the voltage/current at an arbitrary point on a specified transmission line.

3.4 I can use the transmission line equation to match a load to a source using a length of transmission line.

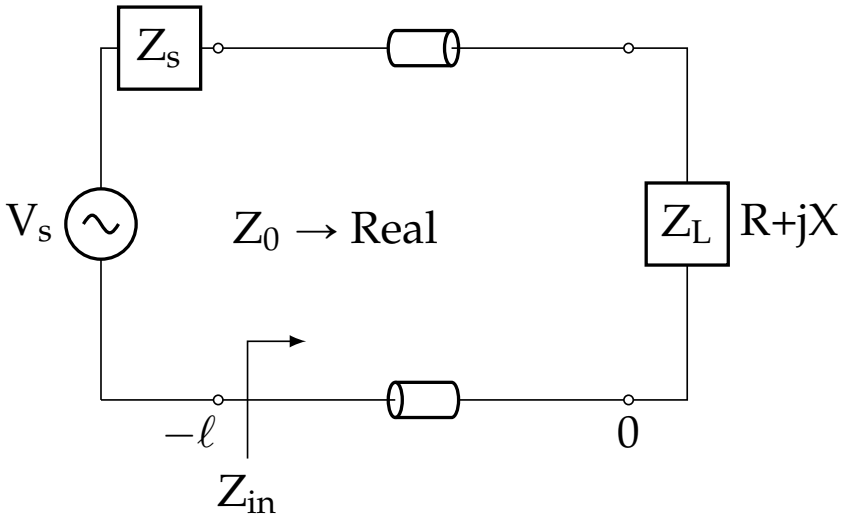
3.5 I can identify the various quantities represented on the Smith Chart.

3.6 I can use a Smith Chart to design a matching network for a given source and load, using series components, parallel components, shorted stubs, and open stubs.

3.7 I can plot the transient response of a load connected to a source through a transmission line.

3.8 I can measure the reflection coefficient across a range of frequencies and interpret the measurement using a network analyzer.

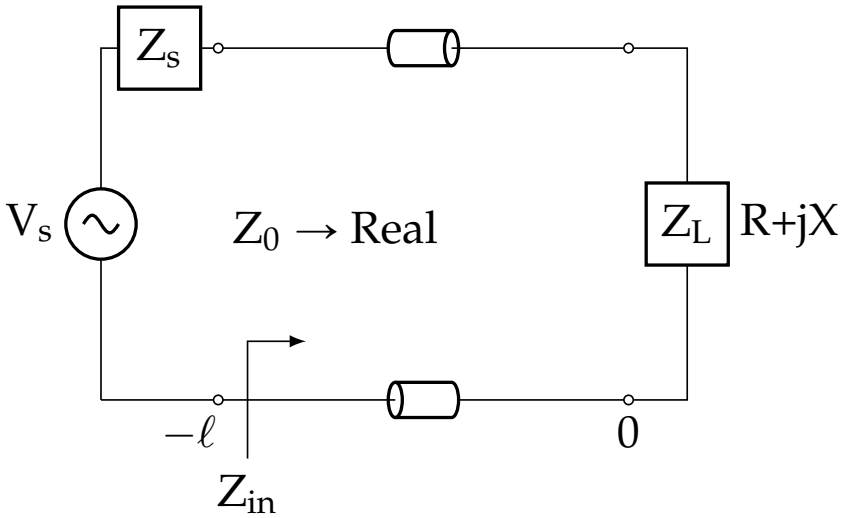
Tx Line Equation



Goal

Match Z_s to Z_{in}, so all P_s is delivered to the load.

Tx Line Equation



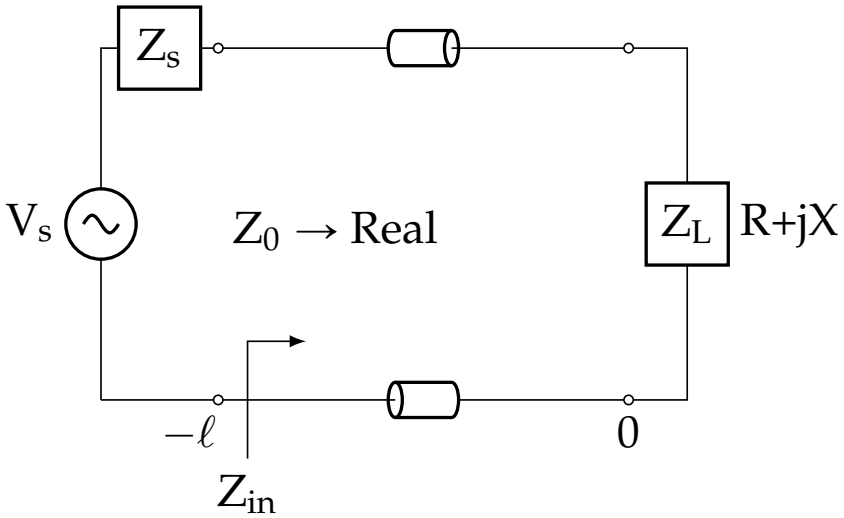
Goal

Match Z_s to Z_{in}, so all P_s is delivered to the load.

Recall the solutions to the voltage and current wave equations:

$$\widetilde{V} = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$
$$\widetilde{I} = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

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With, $\Gamma(z) = |\Gamma_L| e^{2\gamma z} = \frac{V_0^-}{V_0^+} e^{2\gamma z}$, we can write the **lossless case** as:

$$\tilde{V} = V_0^+ e^{-j\beta z} \left(1 + |\Gamma_L| e^{j2\beta z}\right)$$
$$\tilde{I} = I_0^+ e^{-j\beta z} \left(1 + |\Gamma_L| e^{j2\beta z}\right)$$

Tx Line Equation

Now, we can express the **input impedance** (Z_{in}), at $z = -\ell$ as:

$$Z_{in} = \frac{\widetilde{V}}{\widetilde{I}} = \frac{V_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)}{I_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)} = Z_0 \left[\frac{1 + \Gamma_L e^{-j2\beta\ell}}{1 - \Gamma_L e^{-j2\beta\ell}}\right]$$

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$$= Z_0 \left[\frac{1 + \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}{1 - \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}\right]$$

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$$= Z_0 \left[\frac{(Z_L + Z_0)\, e^{+j\beta\ell} + (Z_L - Z_0)\, e^{-j\beta\ell}}{(Z_L + Z_0)\, e^{+j\beta\ell} - (Z_L - Z_0)\, e^{-j\beta\ell}}\right]$$

Tx Line Equation

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$$= Z_0\left[\frac{Z_L 2\cos(\beta\ell) + jZ_0 2\sin(\beta\ell)}{Z_0 2\cos(\beta\ell) + jZ_L 2\sin(\beta\ell)}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan(\beta\ell)}{Z_0 + jZ_L\tan(\beta\ell)}\right]$$

Tx Line Equation

General

$$Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tanh(\gamma \ell)}{Z_0 + jZ_L \tanh(\gamma \ell)} \right]$$

$$Z_{in} = \frac{z_L + j\tanh(\gamma \ell)}{1 + jz_L \tanh(\gamma \ell)}$$

$$z_L = \frac{Z_L}{Z_0}$$

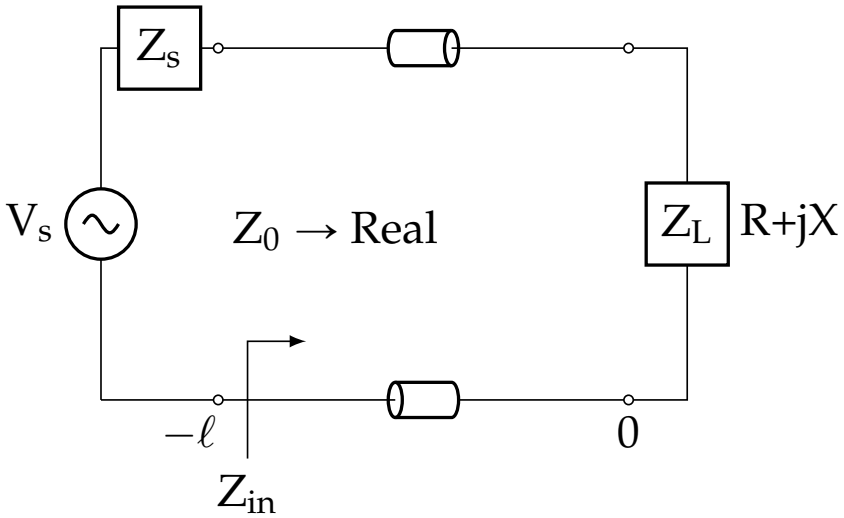
Lossless

$$Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan(\beta \ell)}{Z_0 + jZ_L \tan(\beta \ell)} \right]$$

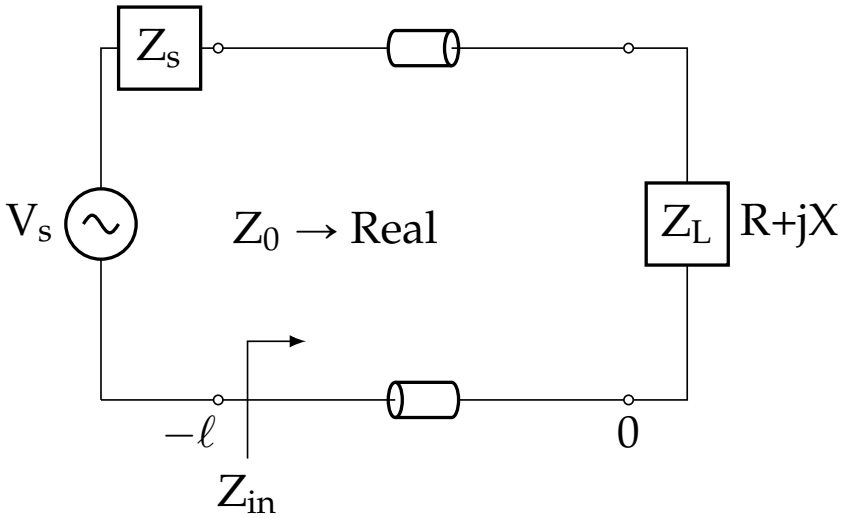
$$Z_{in} = Z_0 \left[\frac{z_L + j\tan(\beta \ell)}{1 + jz_L \tan(\beta \ell)} \right]$$

$$z_L = \frac{Z_L}{Z_0}$$

Tx Line Equation Example: $\ell = \lambda/2$

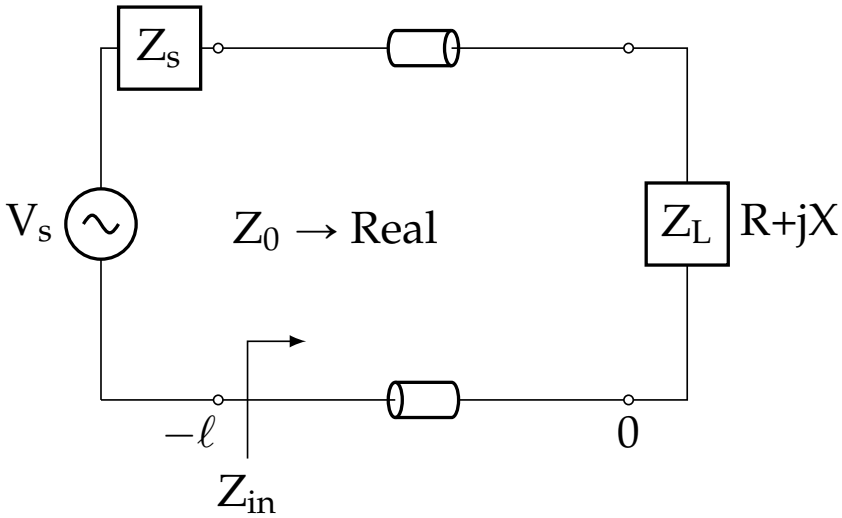


Tx Line Equation Example: $\ell = \lambda/2$



$$\beta = \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi$$

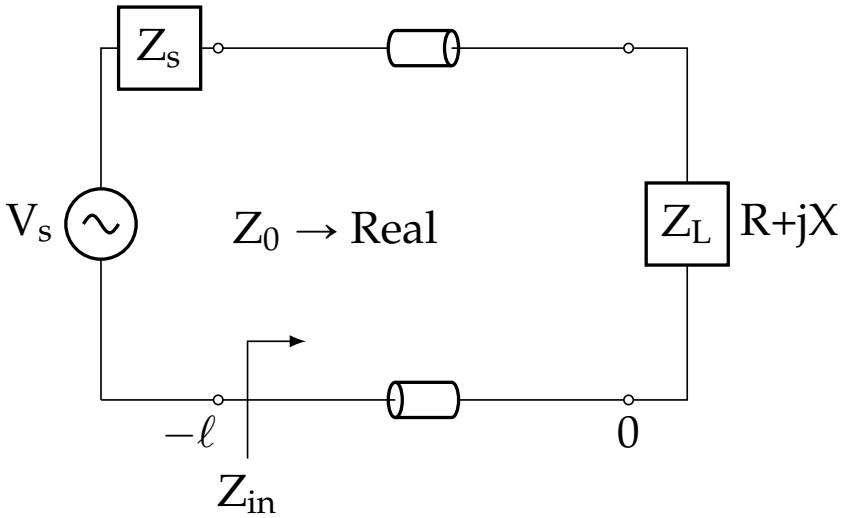
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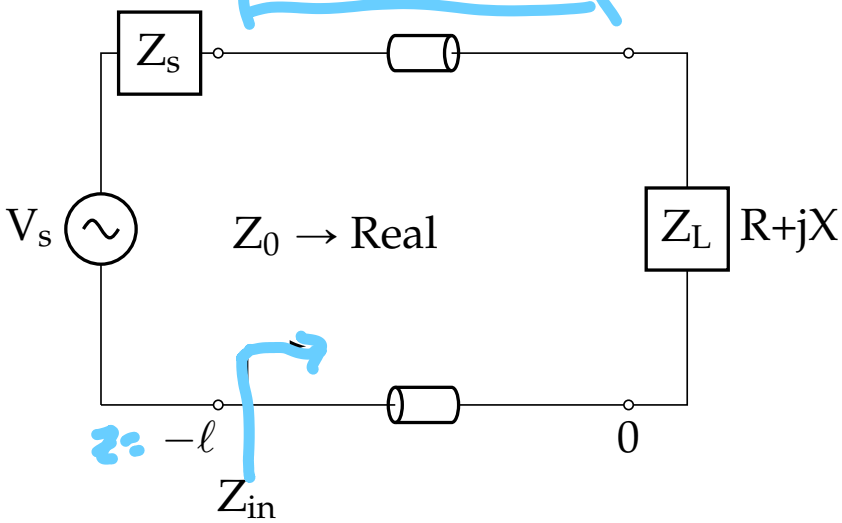
$$= \left[Z_0 \frac{z_L + j\tan(\pi)}{1 + jz_L\tan(\pi)} \right]$$

Tx Line Equation Example: $\ell = \lambda/2$



$$\beta = \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi$$
$$= \left[Z_0 \frac{z_L + j\tan(\pi)}{1 + jz_L\tan(\pi)} \right]$$
$$= \frac{z_L}{1}$$

Tx Line Equation Example: $\ell = \lambda/2$



Half-Wave Transformer

Result

A tx line with a length of $\frac{n\lambda}{2}$ is "transparent".

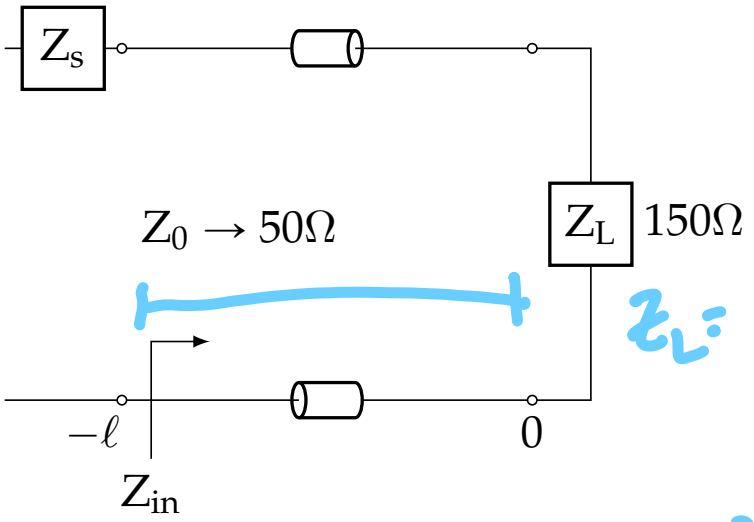
$$\beta = \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi$$

$$Z_{in} = \left[Z_0 \frac{z_L + j\tan(\pi)}{1 + jz_L\tan(\pi)} \right]$$

$$Z_{in} = \frac{z_L}{1}$$

Tx Line Equation: Example #2

Find Γ_L, VSWR, and $Z_{in}(z = -\frac{\lambda}{8})$:



$z_n = \frac{Z_n}{Z_0}$

$z_L = \frac{150}{50} = 3$

1. $\Gamma_L = \frac{150-50}{150+50} = \frac{100}{200} = \frac{1}{2}$

2. VSWR $= \frac{1+|\Gamma_L|}{1-|\Gamma_L|} = \frac{\frac{3}{2}}{\frac{1}{2}} = 3$

3. $Z_{in}(z = -\frac{\lambda}{8})$

$z_{in} = \frac{3 + j\tan(\frac{2\pi}{\lambda}\cdot\frac{\lambda}{8})}{1 + j3\tan(\frac{2\pi}{\lambda}\cdot\frac{\lambda}{8})}$

$= \frac{3+j}{1+j3} = 0.6 - j0.8$

$Z_{in} = z_{in} Z_0$

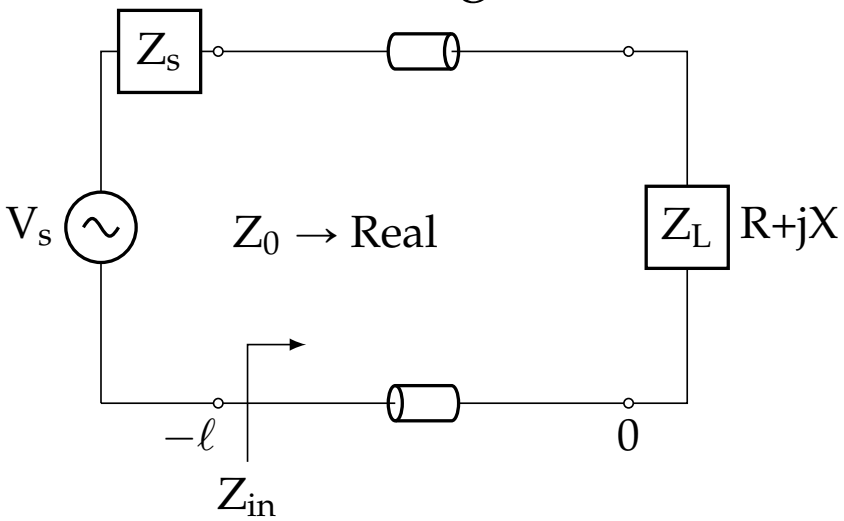
$= (0.6 - j0.8)50\Omega$

$= 30 - j40\ \Omega$

$50\angle -53^\circ\ \Omega$

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



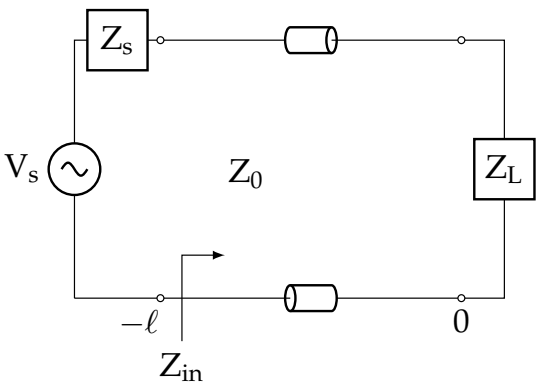
3. $Z_{in}(z = -\frac{\lambda}{8})$

1. Γ_L

2. VSWR

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



Process

1. Normalize impedances to Z_0
2. Find reflection coefficient (Γ_L)
3. Holding the _____ of Γ_L constant, find the input impedance $z_{in} =$ _____

$$\implies |\Gamma_L| = \frac{z_{in} - 1}{z_{in} + 1}$$

4. Determine capacitance or inductance from X.

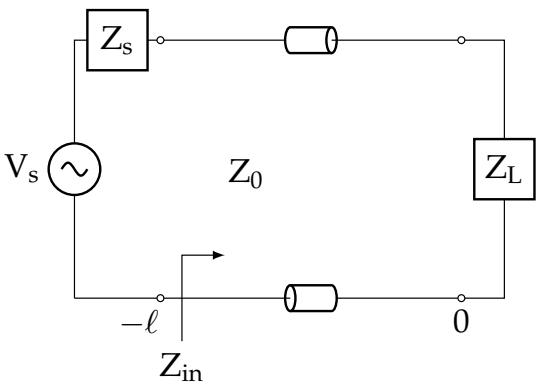
$$\implies Z_C = \frac{1}{j\omega C} \text{ or } Z_L = j\omega L$$

5. Find the length of TX line to make:

$$\implies z_{in} = 1 + jX = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)}$$

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



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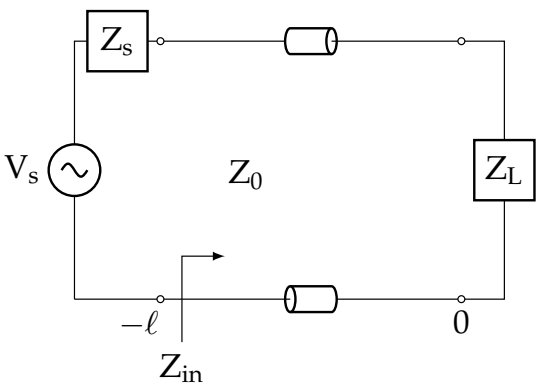
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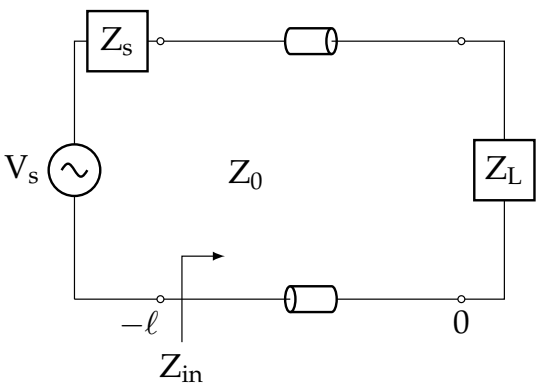
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Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



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2. Find reflection coefficient (Γ_L)
3. Holding the _____ of Γ_L constant, find the input impedance $z_{\text{in}} =$ _____

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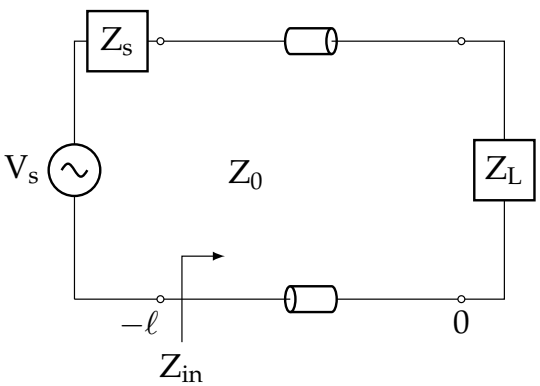
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5. Find the length of TX line to make:

$$\implies z_{\text{in}} = 1 + jX = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)}$$

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



Process

1. Normalize impedances to Z_0
2. Find reflection coefficient (Γ_L)
3. Holding the Mag. of Γ_L constant, find the input impedance $z_{in} =$ Re[z_s] + jX

$$\implies |\Gamma_L| = \frac{z_{in} - 1}{z_{in} + 1}$$

4. Determine capacitance or inductance from X.

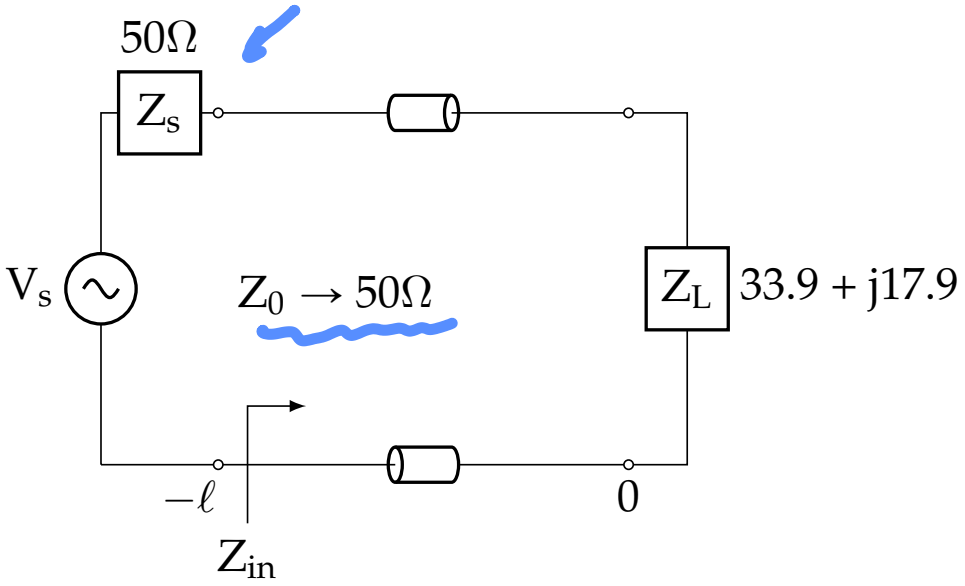
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5. Find the length of TX line to make:

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Impedance Matching: Example

Match the combined load (Z_L and TX line) to Z_s.



1. Normalize Impedances

$z_L = \frac{33.9 + j17.9\ \Omega}{50\ \Omega} = 0.678 + j0.358$

$z_s = 1 \qquad z_0 = 1$

2. Find reflection coefficient

$\Gamma_L = \frac{0.678 + j0.358 - 1}{0.678 + j0.358 + 1}$

$\Gamma_L = -0.141 + j0.239$

$\Gamma_L = 0.278 \angle 120.7°$

Impedance Matching: Example (cont'd)

3. Find z_{in} & C or L

$$|\Gamma_{in}| = |\Gamma_L| = \left|\frac{z_{in}-1}{z_{in}+1}\right| = 0.278$$

Also... $z_{in} = z_s + jx = 1 + jx$

$$0.278 = \left|\frac{1+jx-1}{1+jx+1}\right| = \left|\frac{jx}{2+jx}\right|$$

$$\left(0.278 = \frac{\sqrt{x^2}}{\sqrt{2^2+x^2}}\right)^2$$

$$0.077 = \frac{x^2}{4+x^2}$$

$x = \pm 0.579$ (Negative)

Choose +

$z_{in} = 1 + j0.579$

Need to get rid of $+j0.579$

$$-j0.579 = \frac{1}{j\omega C}$$

$$C = \frac{1}{2\pi f \cdot 0.579 \cdot Z_0}$$

Impedance Matching: Example (cont'd)

4. Determine Tx Line Length ℓ

$$z_{in} = 1 + j0.579 = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)} = \frac{0.678 + j0.358 + j\tan(\beta\ell)}{1 + j(0.678 + j0.358)\tan(\beta\ell)}$$

$$[1 + j0.579][1 + j0.678\tan(\beta\ell) - 0.358\tan(\beta\ell)] = 0.678 + j0.358 + j\tan\beta\ell$$

$$1 + j0.678\tan(\beta\ell) - 0.358\tan(\beta\ell) + j0.579 - j0.207\tan(\beta\ell) - 0.393\tan(\beta\ell) =$$

① collect $A\tan(\beta\ell) + jB\tan(\beta\ell) = C + jD$

② j-terms equal

$$B\tan(\beta\ell) = D$$

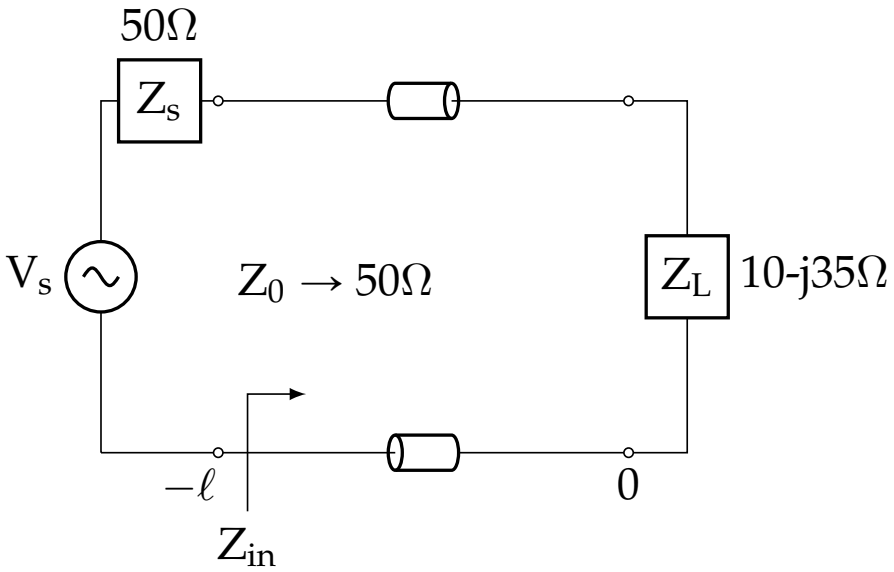
$$\beta\ell = \tan^{-1}\left(\frac{D}{B}\right) = E$$

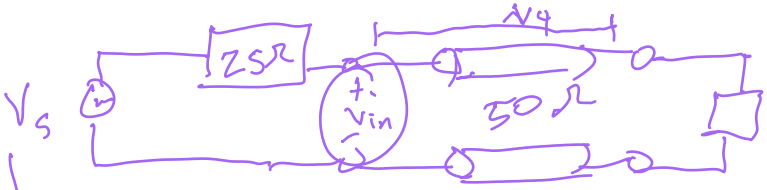
$$\frac{2\pi}{\lambda}\ell = E \Rightarrow \ell = \frac{E}{2\pi}\lambda$$

$\ell = 0.065\lambda$

Impedance Matching: Review

Match the combined load (Z_L and TX line) to Z_s at 1GHz.





Find V_L

$V_s = 10\cos(\omega t + 30°)V$

① $\tilde{V}_s = 10e^{j30°}V = 10\angle 30°\ V$

② $\Gamma_L = \frac{100-50}{100+50} = \frac{1}{3}$

③ Find V_0^+, but first find Z_{in}

$$Z_{in} = \frac{Z_L + j\tan(\beta l)}{1 + jZ_L\tan(\beta l)} = \frac{2 + j\tan(\frac{2\pi}{\lambda}\cdot\frac{\lambda}{4})}{1 + j2\tan(\frac{2\pi}{\lambda}\cdot\frac{\lambda}{4})} =$$

$$Z_{in} = \frac{50^2}{100} = \frac{2500}{100} = 25\Omega$$

④ Voltage divider to find V_{in}

$$V_{in} = V_{ss}\frac{Z_{in}}{Z_{in} + Z_s} = 10\angle 30°\ \frac{25}{50} = 5\angle 30°\ V$$

⑤ Now find V_0^+ -

Quarter Wave Transformer

For $\frac{\lambda}{4}$

$$Z_{in} = \frac{Z_L + j\tan(\frac{\pi}{2})}{1 + jZ_L\tan(\frac{\pi}{2})}$$

$$= \frac{Z_L + j\infty}{1 + jZ_L\infty}\ \frac{\frac{1}{\infty}}{\frac{1}{\infty}}$$

$$= \frac{\frac{Z_L}{\infty} + j}{\frac{1}{\infty} + jZ_L} = \frac{1}{Z_L}$$

$$Z_{in} = \frac{Z_0^2}{Z_L}$$ unitized

$$\tilde{V}\left(z=-\frac{\lambda}{4}\right) = V_0^+\left(e^{+j\frac{2\pi}{\lambda}\cdot\frac{\lambda}{4}} + \Gamma_L e^{-j\frac{\pi}{2}}\right) = 5\angle 30^\circ$$

$\Gamma_L = 1/3$

$$V_0^+ = \frac{5\angle 30^\circ}{1\angle 90^\circ + \frac{1}{3}\angle -90^\circ} = 7.5\angle -60^\circ \text{ V}$$

$$\tilde{V}(z=0) = V_0^+\left(e^{j0} + \Gamma_L e^{j0}\right)$$

$$V_L = 7.5\angle -60^\circ\left(1+\frac{1}{3}\right) = 10\angle -60^\circ \text{ V}$$

$$\boxed{V_L(z,t) = 10\cos(\omega t - 60^\circ)\text{ V}}$$